



Andrea Zingarofalo

Curriculum Vitae & Studiorum

About Me

Theoretical physicist exploring how fundamental physical laws shape the limits of knowledge, communication, and computation. My current research focuses on device-independent methods for certifying quantum resources, a crucial task for establishing trust and security in next-generation quantum technologies.

Education

- 2025 – 2029 **PhD in Quantum Information Theory**, *Université libre de Bruxelles*
 - Thesis: *Self-testing and device-independent verification of quantum computation*
 - Advisor: Dr. S. Pironio
- 2020 – 2025 **Diploma di Licenza**, *Advanced School ISUFI* (Excellence Program), 70/70 cum laude
 - ~10% Acceptance rate for the Natural Sciences Class in 2020
 - Modules: Group Theory and Symmetries, Celestial Mechanics, Astroparticle Physics, Applied Physics I–II, Philosophy of Science, ICT for Research, Public Speaking, Scientific Writing
- 2023 – 2025 **MSc in Theoretical Physics**, *University of Salento*, 110/110 cum laude, GPA: 29.8/30
 - Modules: Quantum Field Theory I–II, QCD, Statistical Mechanics, Quantum Computing, Physics of Complex Systems, Data Analysis, Mathematical Methods for AI
 - Thesis: *Exploring the boundaries of quantum correlations: Bell inequalities, convex duality, and self-testing*
 - Advisors: Prof. L. Martina & Prof. J.-D. Bancal
- 2024 – 2025 **Exchange program**, *École normale supérieure*, Master ICFP (International Center for Fundamental Physics) M2, Theoretical Physics track
 - Modules: General Relativity, Quantum Field Theory, Statistical Field Theory, Algorithms & Computations
- 2020 – 2023 **BSc in Physics**, *University of Salento*, 110/110 cum laude, GPA: 30/30
 - Electives: Nuclear Physics, Astronomy, Foundations of Physics
 - Thesis: *Study of one-dimensional quantum integrable systems via the Bethe Ansatz*
 - Advisor: Prof. L. Martina
- 2015 – 2020 **High School Diploma**, *Liceo Scientifico V. Lilla*, Oria, Italy, 100/100 cum laude
 - Activities: Student ambassador

Experience

- 2025 – 2029 **Doctoral Researcher**, *Laboratoire d'Information Quantique – Université libre de Bruxelles*
 - Research on self-testing and device-independent protocols for certifying components of quantum computation, from multipartite entanglement to error-correcting codes
 - Supervisor: Dr. S. Pironio
- Oct 2025 – Feb 2026 **Visiting PhD Student**, *Institut de Physique Théorique – CEA Paris-Saclay*
 - Research on verification protocols for quantum computation
 - Supervisor: Prof. J.-D. Bancal

- Mar – Jul 2025 **Research Intern**, *Institut de Physique Théorique – CEA Paris-Saclay*
- Derived analytically new Bell inequalities for partially entangled states
 - Implemented SDP relaxations to numerically estimate quantum bounds of Bell inequalities via the Navascués-Pironio-Acín hierarchy
 - Supervisor: Prof. J.-D. Bancal
- Mar – Jul 2024 **Research Intern**, *Mathematical Physics Group – University of Salento (with University of Glasgow)*
- Derived multi-kink solutions of a generalized Fokker-Planck equation via Hirota's bilinear method
 - Developed automated Mathematica scripts for symbolic generation of soliton solutions
 - Supervisor: Prof. L. Martina

Publications

Journal Articles

- [1] F. Giglio, G. Landolfi, L. Martina, and **A. Zingarofalo**, "Integrability properties and multi-kink solutions of a generalised Fokker-Planck equation", *J. Phys. A: Math. Theor.* **58**, 165202 (2025).

Preprints and Work in Preparation

- [1] **A. Zingarofalo**, J.-D. Bancal, and S. Pironio, "Towards photonic Bell experiments with minimal sample complexity", *in preparation* (2026).

Science Communication and Outreach

- [1] **A. Zingarofalo**, "Quantum correlations: A geometric perspective", in *Rete di Idee, Catania 2025*, to be published.
- [2] **A. Zingarofalo**, "Il conflitto tra meccanica quantistica e senso comune", *Il Chiasmo*, Treccani – The Italian Encyclopaedia Institute (2024).
- [3] **A. Zingarofalo**, "Il ruolo delle simmetrie nel problema di Keplero", in *Rete di Idee, Udine 2022*, D. Avaro and F. Decataldo, eds., Forum Editrice, Udine, pp. 81–100 (2023).

Schools and Conferences

Conferences

- 2026 ○ **851. WE-Heraeus-Seminar: 90 Years of EPR**, Physikzentrum Bad Honnef, 7–10 Apr
- Poster: Towards photonic Bell tests with minimal sample complexity
- **VERIQTAS Workshop 2026**, ENS Lyon, 23–25 Mar
- 2025 ○ **RIASISSU "Rete di Idee" XIII**, Scuola Superiore di Catania, Catania, 23 Nov
- Invited Talk: From Bell's theorem to quantum technologies via the device-independent paradigm
- **Workshop in honor of I. Cirac and P. Zoller**, Académie royale de Belgique, Brussels, 3 Oct
- **Interdisciplinary Statistical Physics workshop**, University of Salento, Lecce, 18–19 Sep
- 2022 ○ **RIASISSU "Rete di Idee" X**, Superiore Universitaria di Udine, 9 Oct
- Invited Talk: The role of symmetry in Kepler's problem

Domain-Specific Schools

- 2025 ○ **Qiskit Global Summer School**, IBM, 7–22 Jul
- **Mathematical Aspects of Quantum Information Summer School**, Université Paris-Saclay, 16–20 Jun
- Poster: Orthogonal faces in the CHSH scenario
- 2024 ○ **Theoretical Physics of Fundamental Interactions Summer School**, Scuola Normale Superiore and Scuola Galileiana di Studi Superiori, Pisa, 27–31 May

Others

- 2019 ○ University Orientation School, Scuola Normale Superiore and Accademia Nazionale dei Lincei, Rome, 8–13 Jul

- 2018 ○ Stage at Asiago Astrophysical Observatory, INAF (Italian Institute of Astrophysics) and University of Padua, Asiago, 9–13 Jul
- 2016 ○ Summer School on Astronomy, INAF and SAlt (Italian Astronomical Society), Sant'Agata sui Due Golfi, 5–8 Jul

Awards and Scholarships

- 2025 ○ Selected as one of the few Italian members of Nova – The Global Top Talent Network
- Winner of RIASISSU “Rete di Idee”, XIII edition, Catania
- Selected among the Top 100 STEM graduates in Italy (Class of 2025), by Almalaurea, the national consortium of Italian universities
- Winner of the ISUFI “Research & Innovation” grant for international research mobility (€ 5000)
- 2024 ○ Winner of a Cité Internationale Universitaire de Paris scholarship (accommodation)
- 2022 ○ Winner of RIASISSU “Rete di Idee”, X edition, Udine
- 2020 ○ Winner of the ISUFI Scholarship at the University of Salento, awarded for academic excellence (full accommodation and a stipend)
- 2018 ○ Senior finalist at XVI Italian Astronomy Olympiads, ranking among the top 30/6000 participants

Technical Skills

Programming and Scientific Computing

- Programming Python (NumPy, SciPy, Matplotlib, pandas), C/C++, Julia, Bash; MATHEMATICA
- Quantum Libraries Quantum simulation with Qiskit and PennyLane; SDPs and convex optimization using CVXPY, MOSEK, ncpol2sdpa
- Data Analysis ROOT framework for experimental data, and machine learning basics with scikit-learn and TensorFlow
- Workflow L^AT_EX, Markdown, git version control, Inkscape, Microsoft Office; comfortable with UNIX systems, shell scripting, and basic system management

Communication and Teamwork

Collaborated within international research teams; presented theoretical results at seminars and co-authored peer-reviewed work

Languages

Italian (native), English (C1), French (B2), German (A2)

Memberships

IEEE Quantum Technical Community (2025-), Italian Physical Society – SIF (2023-), RIASISSU Excellence Network (2023-2025)

Additional Information

- Music Guitar Grade 2 (Distinction), Trinity College of London
- Interests Outside academia, I'm a tech enthusiast with a passion for music, history, travelling and geopolitics. I am also an amateur astronomer.